Ashoka University · CS-3410 / MAT-3211-1 · Monsoon 2026

Midterm Project Brief

Biology · Behavioural Economics · Weather and Climate

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This brief is the short version. The full *Midterm Project Guide* has the background, research questions and pitfalls for each track, the evaluation protocol, and the optional SPINN challenge. Where the two seem to differ, the guide gives the detail; the rules, dates, rubric and schedule below are the same tables in both.

1 The rules

Component	Rule
Tracks	Three tracks: **A** Biology (plant stress), **B** Behavioural Economics (risky choice), **C** Weather and Climate (extreme rainfall). Each student chooses *exactly one*. You do not need to attempt the other tracks.
Submission	**Individual.** Every student submits their own code, experiments, report and prediction file, and sits their own viva.
Teams	Teams of three students. **All three members must choose the same track.** A team exists only for the team leaderboard. It does not turn the project into group work. One member registers the team, with all three Kaggle usernames, on the sign-up form `https://forms.gle/1Rovcnb9DzRga1Dj8` (Ashoka Google account) by Tuesday 20 October 2026, 23:59 IST.
Leaderboards	Your individual leaderboard rank earns *no* regular project marks. The **top three teams** on the team leaderboard receive bonus points, shown separately from the 100-point project score. Bonus amounts: 1st team +5, 2nd team +3, 3rd team +2 marks for each member.
Methods	**Classical machine learning only.** Neural networks, deep learning, pretrained models and foundation models are not permitted in the three core tracks (Section 4).

2 Key dates

Event	Date
Project and datasets released (Day 1)	Saturday 17 October 2026
Team and Kaggle-username registration closes	Tuesday 20 October 2026
Final individual submission (Day 14)	Friday 30 October 2026, 23:59 IST
Individual demonstrations and vivas	Monday 2 – Friday 6 November 2026
Leaderboards close (private leaderboard revealed)	Friday 30 October 2026, 23:59 IST

3 The three tracks

All three datasets are **simulated** by the teaching team and released on Kaggle. No row is a real plant, person or weather observation; say so in your report. Each page has `train.csv`, `test.csv` (no labels), `sample_submission.csv` and a data dictionary. A starter notebook per track (`starter_notebooks/`) shows a correct validation split and a valid submission file; it is a protocol, not a good model.

Track	Size (train / test)	How the test set differs from training	Primary metric
A Biology	4,200 / 1,800 plants	Two growing conditions never seen in training, both hotter than any training condition	Macro-averaged F1 (3 classes)
B Behav. Econ.	8,400 / 3,600 decisions	90 participants never seen in training (210 in training, 40 decisions each)	ROC-AUC
C Weather	64,288 / 29,200 station-days	Later years (2021–2024 vs. 2010–2020) and four stations never seen in training	Average precision (PR-AUC)

A Biology: plant stress. Predict `stress_level` (none / moderate / severe) from environmental and physiological measurements; the test set holds two *unseen, hotter* growing conditions. Validate by leaving whole conditions out. Drop `harvest_yield_loss_pct`: it is measured after harvest and is not available at prediction time. Dataset: `https://www.kaggle.com/datasets/samarjeetsinghmalik/iml2026-track-a-simulated-plant-stress`.

B Behavioural Economics: risky choice. Predict the probability that a simulated participant chooses the risky option; gain problems are shown in a gain or a loss *frame*. The test set holds 90 *unseen participants*, so validate with participants grouped (`GroupKFold`). Drop `realised_payout_inr`: it is paid after the choice. Dataset: `https://www.kaggle.com/datasets/samarjeetsinghmalik/iml2026-track-b-simulated-risky-choice`.

C Weather and Climate: extreme rainfall. Predict the probability of ≥ 64.5 mm of rain *tomorrow* at a station. The test set is *later* (2021–2024) and includes *unseen stations*; the climate drifts, so validate chronologically. Drop `rain_tomorrow_mm`: it is the future. Dataset: `https://www.kaggle.com/datasets/samarjeetsinghmalik/iml2026-track-c-simulated-extreme-rain`.

4 What you must do

Methods. Classical machine learning only. Your three or more required baselines are a trivial baseline plus models taught in lectures (regularised linear or logistic regression, LDA/QDA/Naive Bayes, SVM). Untaught classical methods (trees, forests, boosting, k-NN) are allowed if you can explain them in the viva. No neural networks, deep learning, pretrained or foundation models. Generative AI: allowed for coding help and language editing, provided you declare the tool and what you used it for in your README; you must be able to explain every line you submit in the viva, and AI-written analysis must not be presented as your own.

1. One focused, testable research question, and a concise literature review (3–6 sources).
2. At least three baselines, all evaluated with the same protocol.
3. At least two controlled experiments beyond the baseline comparison, each changing one thing.

4. A validation split that matches the test set (conditions, participants or time); all preprocessing fitted inside the training folds; the test file used once.
5. Justified metrics: the track's primary metric plus what your question needs.
6. Error analysis and limitations, including the simulated nature of the data.
7. Reproducibility: fixed seeds, pinned environment, one command that rebuilds your results.

Deliverables (upload to the Google Classroom assignment *IML Midterm Project* by Friday 30 October 2026, 23:59 IST): a 4–6 page report from the template; your notebook or code with a README from the template; your prediction file (also submitted to your track's Kaggle competition); `experimental_results.csv`; and an individual demonstration and viva. Late work: 10% of the project score deducted per day late; not accepted after 2 days; late prediction files do not count for the leaderboard.

5 Assessment

Component	Marks	What earns full marks
Research question and literature review	10	A focused, testable question; three to six relevant sources used, not just listed.
Data understanding, preprocessing and leakage prevention	15	Data audited; preprocessing inside the pipeline; train-only and future columns identified and excluded; a validation split that matches the data's structure.
Baselines and experimental design	20	At least three appropriate baselines; at least two controlled experiments, each testing a stated expectation.
Results, error analysis and limitations	25	Correct metrics with uncertainty where feasible; failure cases examined; honest limitations, including the simulated nature of the data.
Code quality and reproducibility	10	One documented command reproduces the key results and the prediction file; seeds and environment pinned.
Report communication and demonstration	10	A clear 4–6 page report, readable figures and tables, and a working live demonstration.
Individual viva and understanding	10	You can explain and defend every modelling choice without help.
Total	**100**	

Every component is assessed individually.

Team leaderboard (bonus only). Your private-leaderboard score m becomes

$$s = \min\{\max\{(m - b)/(r - b),\, 0\},\, 2\},$$

where b is the track's trivial baseline and r a standard logistic-regression reference; $s = 1$ means "as good as the reference" on every track. A team's score is the mean of its three members' s. Bonus: 1st team +5, 2nd team +3, 3rd team +2 marks for each member. Kaggle limit: 5 submissions per day, with 1 final submission that you nominate. Leaderboard rank earns no regular marks, and upvoting is never assessed. Join your track's competition: A `https://www.kaggle.com/t/c460443624204a52b6c12796be312881`, B `https://www.kaggle.com/t/fc0bc2dabcb04acc9d100158fce95dd5`, C `https://www.kaggle.com/t/aef8156c4b0d4518a7b5c4679f7ad760`.

6 Schedule

Days	Milestone
1–3 (17–19 Oct)	Understand the selected dataset, formulate a research question, review literature, and establish a baseline
4–7 (20–23 Oct)	Implement baselines and conduct initial experiments
8–10 (24–26 Oct)	Run controlled experiments and parameter tuning
11 (27 Oct)	Error analysis and limitations
12–13 (28–29 Oct)	Finalise report, code, reproducibility, and prediction submission
14 (30 Oct)	Final submission; individual demonstrations and vivas follow (2–6 Nov)